# Understand Powers and Exponents

## Be Curious

### Which Doesn't Belong?

$7 \times 7$

$4 \times 4$

$6 \times 8$

$10 \times 10$

Sample explanation

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**Math is... Mindset**

What do you want your classmates to know about your math story?

## Learn

At Week 1, Dean had 10 pennies Each week after, Dean increased the number of pennies by 10 times the previous week.

**During which week will Dean have 1,000,000 pennies?**

You can organize the information in a table to help determine the solution.

**Math is... Patterns**

What patterns do you notice in the table?

| Week | Multiplication Expression | Number of Pennies Added each Week |
|---|---|---|
| 1 | 10 | 10 |
| 2 | $10 \times 10$ | 100 |
| 3 | $10 \times 10 \times 10$ | 1,000 |
| 4 | $10 \times 10 \times 10 \times 10$ | 10,000 |
| 5 | $10 \times 10 \times 10 \times 10 \times 10$ | 100,000 |
| 6 | $10 \times 10 \times 10 \times 10 \times 10 \times 10$ | 1,000,000 |

A **power of 10** is the product of 10 multiplied by itself a number of times.

Dean will have 1,000,000 pennies in Week 6.

You can write a power of 10 as a multplication expression with factors of 10.

You can also write a power of 10 in **exponential form** using a **base** and an **exponent**.

$\underbrace{10 \times 10}_{\text{2 factors}} = 10^{2}$ ← exponent

↑ base

## Work Together

Write $10^{8}$ as a multiplication expression. Then, find the product.

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# On My Own

MATH REPLAY | GO ONLINE

Name _____

**Write the exponential form as a multiplication expression.**

1. $10^4$ $10 \times 10 \times 10 \times 10$
2. $10^2$
3. $10^3$
4. $10^6$

**Write the exponential form.**

5. $10 \times 10 \times 10 =$ _____
6. $10 \times 10 \times 10 \times 10 \times 10 =$ _____
7. $10 \times 10 \times 10 \times 10 =$ _____
8. $10 \times 10 =$ _____

**Write the exponential form of each power of 10.**

9. $10 =$ _____
10. $1{,}000 =$ _____
11. $100 =$ _____
12. $10{,}000 =$ _____

13. Rachel finds the value of $10^5$ as shown. Do you agree with her solution? Tell why.

$10^5 = 10 \times 5 = 50$

14. **STEM Connection** Grace reviewed $10^6$ lines of a computer program. How many lines did she review? Write the product.



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**15.** Trevor's personal walking goal is shown on his activity tracker.




**A.** How can you help Trevor write this goal using an exponent?

**B.** How can you help Trevor write this goal as a product of 10s?

**16.** Jenny's father gives her $10. Her grandfather offers to give her ten times the value her father gave her. Her grandmother offers her a choice of either ten times the value her grandfather offered or $500. Which of her grandmother's offers should Jenny choose? Explain.

$1,000

**17. Extend Your Thinking** Consider the inequality shown.

$$10 \times 10 \times 10 \times 10 < b < 10^6$$

What is the value of $b$? Explain.

## Reflect

What patterns did you notice when writing different forms of powers of 10?

**Math is... Mindset**

How have you told your classmates about your math story?

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